Facing Operation in Turning: A Technical Literature Review

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بسم الله الرحمن الرحيم

دعـــــــــــــــــاء

وَقُلِ اعْمَلُوا فَسَيَرَى اللَّهُ عَمَلَكُمْ وَرَسُولُهُ وَالْمُؤْمِنُونَ

الحمد لله على الإسلام و الايمان

الحمد لله على التمام و الكمال

الحمد لله الذي من علينا فأفضل و أعطانا فأجزل

الحمد لله على السداد و التوفيق و الحمد لله على بلوغ الطريق

" رَبِّ أَوْزِعْنِي أَنْ أَشْكُرَ نِعْمَتَكَ الَّتِي أَنْعَمْتَ عَلَيَّ وَعَلَىٰ وَالِدَيَّ وَأَنْ أَعْمَلَ صَالِحًا تَرْضَاهُ وَأَدْخِلْنِي بِرَحْمَتِكَ فِي عِبَادِكَ الصَّالِحِينَ "

سبحانك اللهم وبحمدك نشهد أن لا إله إلا أنت

نستغفرك و نتوب إليك

وصل اللهم وسلم على سيدنا محمد وعلى آله وصحبه اجمعين

ربنا تقبل منا انك سميع الدعاء

Dedication:

This work is lovingly and wholeheartedly dedicated to our beloved parents, whose endless sacrifices, unconditional love, patience, and continuous encouragement have been the true foundation of our success. Their unwavering belief in our abilities has always inspired us to move forward, overcome every challenge, and strive for excellence. We are deeply grateful for everything they have done for us and for the constant support they have given throughout our academic journey.

We also dedicate this achievement to our families, whose love, understanding, and support have been a source of motivation and strength. To our teachers and mentors, we express our sincere gratitude for their guidance, knowledge, valuable advice, and encouragement, which have greatly contributed to our academic and personal growth.

Finally, we dedicate this work to everyone who supported us, directly or indirectly, throughout this journey. Without their encouragement, guidance, and presence, this accomplishment would not have been possible.

TOUCHI Idriss

Abstract:

Facing is a fundamental turning operation used to produce a flat surface perpendicular to the rotational axis of a workpiece. It is commonly performed at the beginning of a machining sequence to establish a reference face, remove excess material, and obtain the required axial dimension. In conventional and CNC turning, the operation is performed by feeding a single-point cutting tool radially across the end face of a rotating workpiece. Facing is therefore closely related to the machine-tool kinematics, cutting-tool geometry, cutting conditions, and resulting surface quality. Standardized terminology for cutting-tool geometry and tool/workpiece motion is defined within the ISO 3002 series.

Research on turning has demonstrated that machining parameters such as cutting speed, feed rate, and depth of cut can significantly influence cutting forces and surface roughness. However, the magnitude and direction of these effects depend on the workpiece material, cutting-tool material and geometry, machine-tool characteristics, and machining conditions.

This review presents the fundamentals of facing, its operating principle, tool requirements, cutting parameters, surface generation, common defects, CNC implementation, and its relationship with machining performance and quality.

Keywords: Facing, Turning, Lathe, CNC Turning, Machining Parameters, Cutting Tool, Surface Roughness, Cutting Forces, Machining Operations.

General Introduction:

Turning is one of the most widely used machining processes in modern manufacturing, particularly for producing cylindrical and rotationally symmetric components. It is a material-removal process in which a cutting tool removes material from a rotating workpiece to achieve the required geometry, dimensions, and surface characteristics. Owing to its versatility, turning is extensively applied in the production of components used in mechanical, automotive, aerospace, energy, and general industrial applications.

Among the various operations performed on a lathe, **facing** is one of the fundamental and frequently performed operations. Facing consists of feeding a cutting tool across the end surface of a rotating workpiece in order to generate a flat surface approximately perpendicular to the axis of rotation. Although the operation is relatively simple in principle, its performance directly influences dimensional accuracy, surface quality, and the preparation of the workpiece for subsequent machining operations.

The quality and efficiency of a facing operation depend on several interacting factors, including cutting speed, feed rate, depth of cut, cutting-tool geometry, workpiece material, machine-tool rigidity, and the use of cutting fluids. These parameters can influence important machining responses such as cutting forces, cutting temperature, tool wear, material removal rate, and surface roughness. Consequently, appropriate selection and control of machining conditions are essential for achieving the desired balance between productivity, tool life, dimensional accuracy, and surface quality.

The development of **Computer Numerical Control (CNC)** technology has further expanded the capabilities of facing operations by enabling precise and repeatable control of tool motion and machining parameters. Modern CNC turning systems can integrate facing with other operations such as longitudinal turning, grooving, drilling, boring, and threading within a single machining process, making effective process planning and parameter selection increasingly important. This paper presents a technical review of the facing operation in turning manufacturing. It discusses its fundamental principles, machining kinematics, cutting tools, process parameters, surface generation, cutting forces, tool wear, surface roughness, CNC implementation, and common machining problems. Particular attention is given to the relationships between machining conditions and process performance, with the aim of providing a structured understanding of facing from both manufacturing and research perspectives.

turning operations (Facing)

1.Introduction:

Turning is one of the principal material-removal processes used in the manufacture of rotational components. A wide range of operations can be performed on a lathe, including straight turning, taper turning, profiling, facing, grooving, boring, threading, drilling, and parting. Kalpakjian and Schmid classify facing among the principal operations performed on a lathe.

Among these operations, facing is particularly important because it produces a planar surface at the end of a cylindrical workpiece. In a conventional description of the operation, the cutting tool is fed in a radial direction across the rotating workpiece, thereby removing material from the end face. Groover similarly describes facing as an operation in which the tool is fed radially to produce a surface perpendicular to the axis of rotation.

Facing is not necessarily performed only to obtain the final surface finish. It can also establish a reference surface from which subsequent dimensions are measured. Consequently, the quality and accuracy of a facing operation can influence subsequent machining operations and the dimensional accuracy of the finished component.

2.Definition and Principle of Facing:

Facing can be defined as a turning operation in which material is removed from the end of a rotating workpiece to generate a planar surface approximately perpendicular to its axis of rotation.

The fundamental kinematic principle can be represented as:

Workpiece → rotational motion

Cutting tool → radial feed motion

The combination of these motions produces the machined face.

The operation is illustrated among the standard lathe operations in manufacturing-process literature, where facing is distinguished from longitudinal turning by the direction of tool feed.

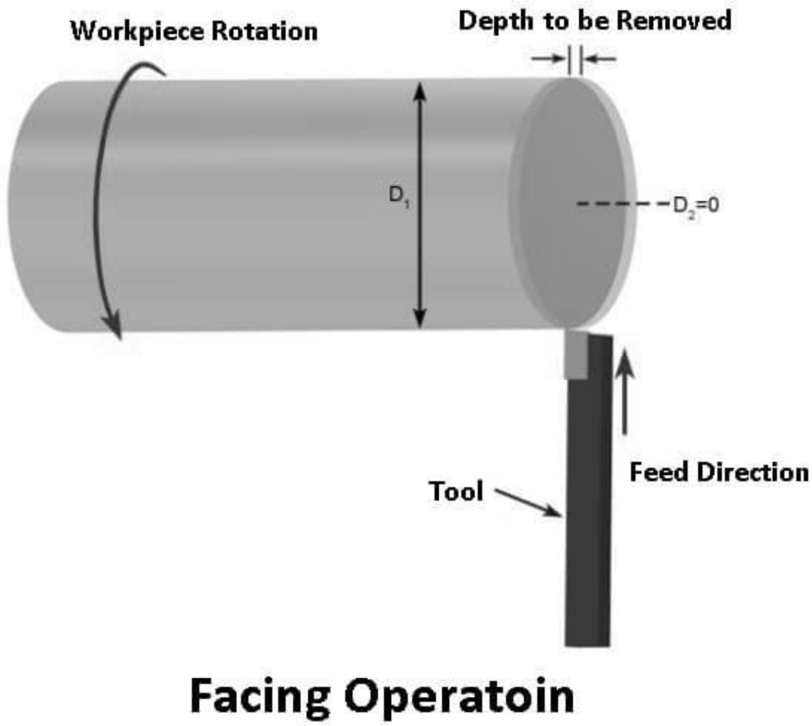


Figure 1. Basic principle of facing

3.Purpose of the Facing Operation:

Facing may serve several manufacturing purposes:

1. **Establishing a reference surface:**

A flat end face can provide a datum or reference from which axial dimensions are subsequently established.

3.1. Removing excess material

Facing can remove irregular or excess material from the end of raw stock before subsequent operations.

3.2. Obtaining the required axial dimension

Material can be removed until the workpiece reaches the required length.

3.3. Improving surface condition

The operation can replace an irregular or rough initial end surface with a controlled machined surface.

3.4. Preparing the workpiece for subsequent operations

Facing is frequently incorporated into a machining sequence before operations such as drilling, boring, threading, or turning to final dimensions. Manufacturing laboratory curricula, for example, commonly teach facing alongside step turning, taper turning, undercutting and threading on conventional and CNC lathes.

4.Facing on Conventional and CNC Lathes:

Facing can be performed on both conventional and CNC turning machines:

In a conventional lathe, the operator controls the tool movement manually through the machine's feed mechanisms.

In a CNC lathe, the same fundamental machining principle is implemented through programmed tool motion. CNC control allows the machine to execute predefined tool paths and machining parameters with greater repeatability.

Experimental research comparing conventional and CNC turning has reported differences in resulting surface roughness under otherwise similar machining conditions, demonstrating that machine-tool characteristics can influence the final surface quality.

This distinction is important because facing performance cannot be evaluated solely from cutting parameters; the machine tool itself is also part of the machining system.

5.Cutting Tool and Tool Geometry:

Facing is generally performed using a single-point turning tool or an indexable insert system appropriate for the workpiece material and machining condition.

Tool geometry affects:

- cutting forces
- chip formation
- surface generation
- tool wear
- achievable surface finish

The terminology and reference systems used to describe cutting-tool geometry are standardized by ISO 3002-1, which defines general terms relating to tool surfaces, cutting edges, tool/workpiece motions, reference planes, and tool and working angles.

Important geometric characteristics include:

- rake angle;
- clearance angle;
- cutting-edge angles;
- nose radius;
- insert geometry.

Tool geometry therefore needs to be considered together with machining parameters rather than as an isolated factor.

6.Cutting Parameters in Facing:

The principal machining parameters relevant to facing include:

6.1.Cutting speed

Cutting speed represents the relative velocity between the cutting edge and the workpiece at the cutting zone.

For turning:

$Vc=1000\pi DN$

where:

- Vc = cutting speed (m/min)
- D = workpiece diameter (mm)
- N = spindle speed (rev/min)

One particular characteristic of facing is that the cutting diameter changes as the tool moves toward the center. Therefore, under constant spindle speed, the instantaneous cutting speed changes during the operation.

This is one reason why CNC turning systems can employ constant surface-speed control when appropriate.

6.2.Feed rate

Feed determines how rapidly the cutting tool advances across the workpiece.

Depending on the CNC programming convention, feed may be specified in (mm/rev) or (mm/min)

Feed is particularly important for surface generation.

Experimental turning studies have repeatedly identified feed rate as an important factor affecting surface roughness and cutting forces. For example, Asiltürk and Akkuş found feed rate to be the most significant factor for Ra and Rz in their hard-turning experiments using Taguchi analysis.

6.3.Depth of cut

Depth of cut represents the amount of material removed during a pass.

In facing, the actual cutting engagement changes with the radial position of the tool, so the process should not be interpreted exactly like constant-diameter longitudinal turning.

Studies of turning have shown that depth of cut can have a significant effect on cutting force, although its influence on surface roughness may be smaller than that of feed under particular experimental conditions.

7.Surface Roughness in Facing:

Surface roughness is one of the principal indicators used to evaluate machining quality.
The final surface is influenced by several interacting variables:

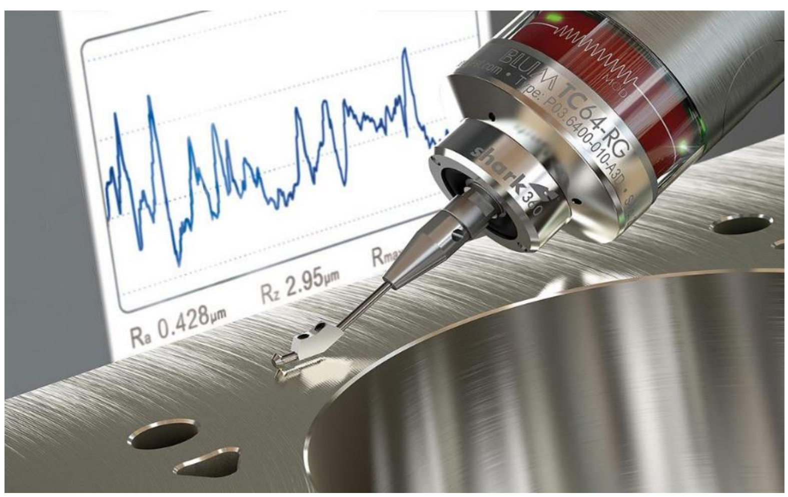


Figure 2. Surface Roughness in Facing

It is therefore inappropriate to attribute surface roughness to one machining parameter alone.

Experimental studies in turning have demonstrated the importance of cutting conditions on surface roughness. Al-Dolaimy reported that cutting speed, feed, depth of cut, cutting fluid, and machine-tool type affected the measured surface roughness in turned steel components.

Likewise, Asiltürk and Akkuş demonstrated statistically that feed rate was the dominant factor for Ra and Rz in their investigated hard-turning conditions.

These results should not be generalized to every facing condition, because the experiments concern specific materials, tools, machines, and parameter ranges.

8.Cutting Forces

During facing, the interaction between the cutting edge and workpiece generates cutting forces.

ISO 3002-4 establishes terminology relating to forces, torques, energy, and power associated with cutting operations.

Cutting forces are important because excessive forces can contribute to:

- tool deflection
- workpiece deflection
- dimensional errors
- vibration
- increased energy consumption
- tool wear.

Experimental research in turning has shown that feed rate and depth of cut can strongly influence cutting forces. In one Taguchi-based investigation, feed rate significantly influenced cutting force, while depth of cut also had a significant influence on cutting force.

Research specifically involving facing has also used facing experiments to investigate cutting-force and tool-wear behavior under varying cutting speed and workpiece hardness.

9.Tool Wear During Facing

Tool wear is another important factor affecting facing performance.

Common wear mechanisms in turning include:

- flank wear
- crater wear

- edge chipping
- notch wear.

As the cutting edge wears, the geometry of the tool changes, which can subsequently influence cutting forces and surface texture.

Research on turning has demonstrated the relationship between machining parameters, tool wear, cutting forces and surface roughness. For example, studies of hard turning have simultaneously investigated cutting forces, tool wear and surface roughness under different machining conditions.

Thus, tool condition should be considered when evaluating the quality of a facing operation.

10.Facing and Cutting-Fluid Application

Cutting fluids can influence machining performance through cooling and lubrication.
Their potential effects include:

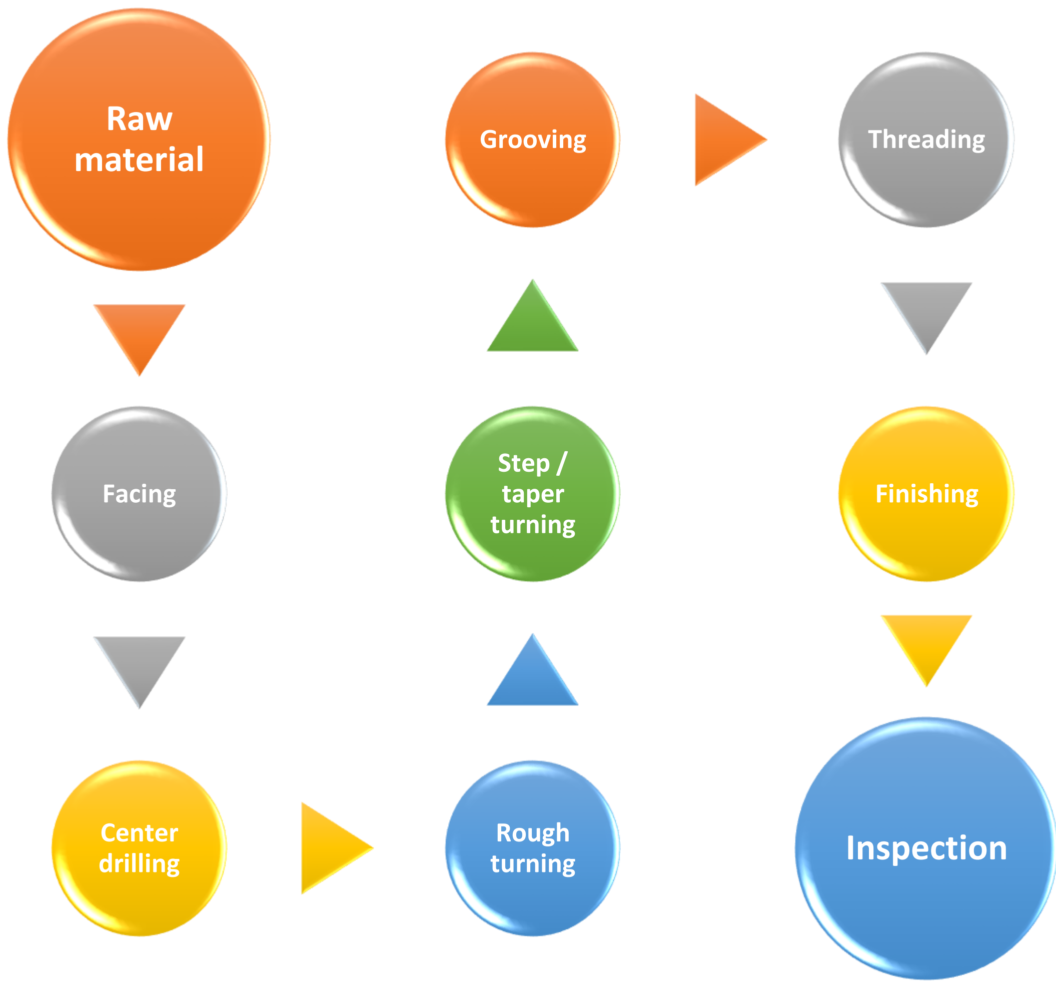
- reducing temperature;
- reducing friction;
- improving tool life;
- influencing surface roughness;
- improving chip evacuation.

Experimental work in turning has found that cutting-fluid use can influence surface roughness.

More recent research has also investigated optimization of cutting-fluid concentration together with machining parameters, demonstrating that fluid conditions and cutting parameters can interact in determining surface roughness and tool flank wear.

11.Facing in CNC Manufacturing

In CNC manufacturing, facing is commonly incorporated into the machining sequence before other operations.
A simplified process plan may be:



This sequence is not universal; the appropriate sequence depends on the component geometry, work holding method, dimensional requirements, and manufacturing strategy.

Facing can therefore serve as an important first machining operation for establishing a controlled reference surface.

12.Research Findings and Comparative Analysis

The literature indicates that facing and turning performance is controlled by an interaction of several variables rather than a single parameter.

Factor	Potential influence
Cutting speed	Temperature, tool wear, surface finish
Feed rate	Surface roughness, cutting forces
Depth of cut	Cutting forces, material removal
Tool geometry	Forces, chip formation, surface quality
Workpiece material	Machinability, forces, tool wear
Tool material	Tool life, allowable cutting conditions
Coolant	Temperature, friction, surface quality
Machine rigidity	Vibration, dimensional accuracy

For example, one experimental investigation found feed rate to have the most significant influence on both cutting force and surface roughness, while depth of cut had a significant influence on cutting force but not surface roughness under the investigated conditions.

Another study on hard turning found feed rate to be the dominant factor affecting Ra and Rz.

These findings demonstrate why machining optimization must be based on the **specific workpiece–tool–machine system** rather than universal parameter rules.

13.Research Gaps

The literature shows that considerable research has been devoted to optimizing turning parameters, surface roughness, cutting forces and tool wear. However, results obtained under one combination of workpiece material, tool geometry, machine and cutting conditions cannot necessarily be transferred directly to another machining system.

Future research can therefore focus on:

- multi-objective optimization
- real-time monitoring of tool condition
- AI-based prediction of surface quality
- energy-efficient facing
- sustainable cutting fluids
- digital monitoring of CNC machining

simultaneous optimization of surface roughness, productivity and tool life.

Recent work in turning has already begun combining machining data with machine-learning models to predict and optimize multiple machining responses, illustrating the movement toward data-driven machining.

14.Conclusion:

Facing is a fundamental turning operation in which a cutting tool is fed radially across the end of a rotating workpiece to produce a planar surface. Its apparent simplicity hides a complex interaction among machine-tool kinematics, cutting-tool geometry, cutting parameters, workpiece material, tool wear, and thermal and mechanical phenomena.

The reviewed literature demonstrates that feed rate, cutting speed, depth of cut and other process variables can significantly influence cutting forces and surface quality, but their effects depend strongly on the specific machining conditions.

Consequently, effective facing should be treated as an integrated manufacturing process rather than simply a preliminary operation. In modern CNC manufacturing, optimization of facing conditions can contribute to dimensional accuracy, surface quality, productivity and tool utilization.

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